

Fig. 4 Audit map — 14 contrast-units from 12 public GEO series

	Batch provenance	Groups in a single batch	Lead PC R ² (batch)	Audit verdict
POCD / PND	GSE330865 GSM block	6	0.26	caveat
	GSE297195 GSM block	6	0.99	caveat
	GSE283401 inferred	13	0.05	caveat
	GSE304956 measured	0	0.02	unusable
	GSE276942 design	5	0.36	unusable
	GSE163943 none	0	0.00	caveat
TBI / neurodeg.	GSE59645 measured	6	0.96	unusable
	GSE115614 measured	5	0.82	caveat
	GSE31357 inferred	7	0.01	caveat
	GSE68207 inferred	0	0.81	unusable
	GSE64978 inferred	0	0.04	unusable
	GSE80174 ctx inferred	0	0.16	usable
	GSE80174 thal inferred	0	0.00	usable
	GSE80174 hippo inferred	1	0.67	unusable

Measured batch = scan date / scanner / declared processing block. GSM block = accession-run blocks (submitter batch column UNKNOWN). Inferred = median split of library size, no bimodality. The verdict attaches to a contrast-unit, not to a GEO series.